Evaluation runs: 2pc-le (election)

generated 2026-11-03 from brain-data e6c29d7dd 2026-11-03 20:14:23 +0100

Every campaign run on this benchmark, one row per instance (variant, specification, communication model, size) and engine; the newest run of an instance wins. Expected verdicts (the rule file): leader-conf: expected conforms; ring-conf: expected conforms; leader-silent: expected violation; leader-split: expected violation; leader-split: expected conforms (one participant: nothing to split); ring-split: expected violation; ring-split: expected conforms (one participant: nothing to split). Blue rows are at the current pins (TraceForge 367cdc6, whose predecessor 9a0629e is equivalent for every engine but the join; merge join 367cdc6; JMC 3c43b2d2): a rerun gives the same result. Red rows are superseded (an older pin, or the in-memory join that the merge join replaces) and their rerun is pending. A planned instance without a measurement is not a row: it is listed under "Not run". G_{spec}, G_{impl} = the graphs of the specification and of the implementation (the stateless engine's own count of specification graphs explored, repeats included, is G_{spec}^{expl}); G_x = the implementation graphs no specification graph covers (violating pairs); T = wall seconds; M = peak resident MiB; tier = the run's time and memory limits; * marks a verdict other than the expected one; 🕓 = the time tier was reached; oom = the memory limit was reached.

stateless

stateless: pair leader-conf, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	sweeps	G_{spec}^{expl}	run
2	4	6	conforms	0.0	4	1 h, 250 GB	2	3	brain03, 2026-11-03
3	18	588	conforms	0.1	4	1 h, 250 GB	6	93	brain03, 2026-11-03
4	96	1839744	conforms	22.9	22	1 h, 250 GB	24	300	brain03, 2026-11-03
5			🕓 (1 h)			1 h, 250 GB			brain03, 2026-11-03

stateless: pair ring-conf, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	sweeps	G_{spec}^{expl}	run
2	4	4	conforms	0.0	4	1 h, 250 GB	2	7	brain03, 2026-11-03
3	18	24	conforms	0.0	2	1 h, 250 GB	6	21	brain03, 2026-11-03
4	96	192	conforms	0.1	4	1 h, 250 GB	24	1452	brain03, 2026-11-03
5	600	1920	conforms	0.8	6	1 h, 250 GB	120	7260	brain03, 2026-11-03
6	4320	23040	conforms	104.0	14	1 h, 250 GB	720	1814760	brain03, 2026-11-03
7	35280	322560	conforms	1760.5	30	1 h, 250 GB	5285	66628084	brain03, 2026-11-03
8	322560		🕓 (1 h)			1 h, 250 GB			brain03, 2026-11-03

stateless: pair leader-silent, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	sweeps	G_{spec}^{expl}	run
2	4	3	3	violation	0.0	4	1 h, 250 GB	3	12	brain03, 2026-11-03
3	18	196	196	violation	0.2	4	1 h, 250 GB	196	3528	brain03, 2026-11-03
4	96	459936	459936	violation	422.1	23	1 h, 250 GB	459936	44153856	brain03, 2026-11-03
5				🕓 (1 h)			1 h, 250 GB			brain03, 2026-11-03

stateless: pair leader-split, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	sweeps	G_{spec}^{expl}	run
2	4	9	9	violation	0.0	4	1 h, 250 GB	9	36	brain03, 2026-11-03
3	18	882	882	violation	0.8	4	1 h, 250 GB	882	15876	brain03, 2026-11-03
4	96	2759616	2759616	violation	2213.8	28	1 h, 250 GB	2759616	264923136	brain03, 2026-11-03

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	sweeps	G^{expl}_{spec}	run
5				(1 h)			1 h, 250 GB			brain03, 2026-11-03

stateless: pair **ring-split**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	sweeps	G^{expl}_{spec}	run
2	4	6	6	violation	0.0	4	1 h, 250 GB	6	24	brain03, 2026-11-03
3	18	28	28	violation	0.1	4	1 h, 250 GB	28	504	brain03, 2026-11-03
4	96	120	120	violation	0.7	6	1 h, 250 GB	120	11520	brain03, 2026-11-03
5	600	496	496	violation	19.0	12	1 h, 250 GB	496	297600	brain03, 2026-11-03
6	4320	2016	2016	violation	543.1	17	1 h, 250 GB	2016	8709120	brain03, 2026-11-03
7	35280			(1 h)			1 h, 250 GB			brain03, 2026-11-03

Not run (stateless). skipped: a smaller size had reached its limit: leader split 6 fifo, leader conf 6 fifo, leader silent 6 fifo, leader conf 7 fifo, leader silent 7 fifo, leader split 7 fifo, leader silent 8 fifo, leader conf 8 fifo, ring split 8 fifo, leader split 8 fifo, ring split 9 fifo, leader split 9 fifo, leader conf 9 fifo, leader silent 9 fifo, ring conf 9 fifo, ring split 10 fifo, ring conf 10 fifo.

planned but never scheduled (grid points beyond the ladder): ring split 9 fifo, ring conf 9 fifo, ring conf 10 fifo, ring split 10 fifo, ring conf 11 fifo, ring split 11 fifo.

stateful

stateful: pair **leader-conf**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	run
2	4	6	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-01
3	18	588	conforms	0.1	4	1 h, 250 GB	brain02, 2026-11-01
4	96	1839744	conforms	12.3	4	1 h, 250 GB	brain02, 2026-11-01
5			(1 h)			1 h, 250 GB	brain02, 2026-11-03

stateful: pair **ring-conf**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	run
2	4	4	conforms	0.0	2	1 h, 250 GB	brain02, 2026-11-03
3	18	24	conforms	0.0	2	1 h, 250 GB	brain02, 2026-11-03
4	96	192	conforms	0.1	2	1 h, 250 GB	brain02, 2026-11-03
5	600	1920	conforms	0.4	2	1 h, 250 GB	brain02, 2026-11-03
6	4320	23040	conforms	4.4	6	1 h, 250 GB	brain02, 2026-11-03
7	35280	322560	conforms	33.8	38	1 h, 250 GB	brain02, 2026-11-03
8	322560	5160960	conforms	288.2	151	1 h, 250 GB	brain02, 2026-11-03
9	3265920	92897280	conforms	1893.5	1438	48 h, 250 GB	brain05, 2026-11-02
10	36288000	1857945600	conforms	45668.0	19992	48 h, 250 GB	brain05, 2026-11-02
12			oom (250 GB)			48 h, 250 GB	brain05, 2026-11-03

stateful: pair **leader-silent**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	run
2	4	3	3	violation	0.0	2	1 h, 250 GB	brain02, 2026-11-03
3	18	196	196	violation	0.0	2	1 h, 250 GB	brain02, 2026-11-03
4	96	459936	459936	violation	9.2	4	1 h, 250 GB	brain02, 2026-11-03
5				(1 h)			1 h, 250 GB	brain02, 2026-11-03

stateful: pair **leader-split**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	run
2	4	9	9	violation	0.0	2	1 h, 250 GB	brain02, 2026-11-03
3	18	882	882	violation	0.1	2	1 h, 250 GB	brain02, 2026-11-03

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	run
4	96	2759616	2759616	violation	39.5	24	1 h, 250 GB	brain02, 2026-11-03
5				(1 h)			1 h, 250 GB	brain02, 2026-11-03

stateful: pair **ring-split**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	run
2	4	6	6	violation	0.0	2	1 h, 250 GB	brain02, 2026-11-03
3	18	28	28	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-03
4	96	120	120	violation	0.1	2	1 h, 250 GB	brain02, 2026-11-03
5	600	496	496	violation	0.2	4	1 h, 250 GB	brain02, 2026-11-03
6	4320	2016	2016	violation	1.1	6	1 h, 250 GB	brain02, 2026-11-03
7	35280	8128	8128	violation	4.8	12	1 h, 250 GB	brain02, 2026-11-03
8	322560	32640	32640	violation	31.4	147	1 h, 250 GB	brain02, 2026-11-03
9	3265920	130816	130816	violation	157.7	1313	48 h, 250 GB	brain05, 2026-11-02
10	36288000	523776	523776	violation	1610.0	17885	48 h, 250 GB	brain05, 2026-11-02

Not run (stateful). planned but never scheduled (grid points beyond the ladder): ring conf 11 fifo, ring split 11 fifo, ring split 12 fifo, ring split 13 fifo, ring conf 13 fifo.

skipped: a smaller size had reached its limit: leader split 6 fifo, leader conf 6 fifo, leader silent 6 fifo, leader conf 7 fifo, leader silent 7 fifo, leader split 7 fifo, leader silent 8 fifo, leader conf 8 fifo, leader split 8 fifo, leader split 9 fifo, leader conf 9 fifo, leader silent 9 fifo.

left out by the table rules (ok): ring split 9 fifo, ring conf 9 fifo, ring split 10 fifo.

left out by the table rules (timeout): ring conf 10 fifo.

offline

offline: pair **leader-conf**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	4	6	conforms	0.1	4	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
3	18	588	conforms	0.1	4	1 h, 250 GB	0.1	0.1	0.0	merge	brain05, 2026-11-03
4	96	1839744	conforms	16.3	6	1 h, 250 GB	0.1	16.3	0.0	merge	brain05, 2026-11-03

offline: pair **ring-conf**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	4	4	conforms	0.1	4	10 min, 16 GB	0.1	0.0	0.0	in-memory	brain05, 2026-11-02
3	18	24	conforms	0.1	4	10 min, 16 GB	0.1	0.0	0.0	in-memory	brain05, 2026-11-02
4	96	192	conforms	0.1	4	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
5	600	1920	conforms	0.3	6	10 min, 16 GB	0.1	0.3	0.0	in-memory	brain05, 2026-11-02
6	4320	23040	conforms	3.0	10	1 h, 250 GB	0.5	2.7	0.3	merge	brain05, 2026-11-03
7	35280	322560	conforms	22.4	67	1 h, 250 GB	3.3	19.7	2.7	merge	brain05, 2026-11-03
8	322560	5160960	conforms	123.5	1155	1 h, 250 GB	26.9	95.5	28.0	merge	brain05, 2026-11-03
9	3265920	92897280	conforms	2173.4	11659	1 h, 250 GB	279.6	1857.8	315.6	merge	brain05, 2026-11-03

offline: pair **leader-silent**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	4	3	1	violation	0.1	4	10 min, 16 GB	0.1	0.0	0.0	in-memory	brain05, 2026-11-02
3	18	196	2	violation	0.1	4	1 h, 250 GB	0.1	0.1	0.0	merge	brain05, 2026-11-03
4	96	459936	6	violation	6.2	6	1 h, 250 GB	0.1	6.2	0.0	merge	brain05, 2026-11-03

offline: pair **leader-split**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	4	9	3	violation	0.1	4	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
3	18	882	9	violation	0.2	4	1 h, 250 GB	0.1	0.2	0.0	merge	brain05, 2026-11-03
4	96	2759616	36	violation	30.3	8	1 h, 250 GB	0.1	30.3	0.0	merge	brain05, 2026-11-03

offline: pair **ring-split**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	4	6	3	violation	0.1	4	10 min, 16 GB	0.1	0.0	0.0	in-memory	brain05, 2026-11-02
3	18	28	7	violation	0.1	4	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
4	96	120	15	violation	0.1	4	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
5	600	496	31	violation	0.2	6	10 min, 16 GB	0.1	0.2	0.0	in-memory	brain05, 2026-11-02
6	4320	2016	63	violation	0.8	10	1 h, 250 GB	0.5	0.5	0.2	merge	brain05, 2026-11-03
7	35280	8128	127	violation	5.7	67	1 h, 250 GB	3.3	1.6	2.3	merge	brain05, 2026-11-03
8	322560	32640	255	violation	51.9	1155	1 h, 250 GB	26.9	5.1	25.0	merge	brain05, 2026-11-03
9	3265920	130816	511	violation	565.1	11659	1 h, 250 GB	279.6	15.1	285.5	merge	brain05, 2026-11-03
10	36288000	523776		(1 h)			1 h, 250 GB	2761.2	41.4		merge	brain05, 2026-11-03

Not run (offline). planned but never scheduled (grid points beyond the ladder): ring split 9 fifo, ring conf 9 fifo, ring conf 10 fifo, ring split 10 fifo, ring conf 11 fifo, ring split 11 fifo, ring split 12 fifo.

left out by the table rules (ok): leader conf 4 fifo, ring split 4 fifo, ring conf 4 fifo, leader silent 4 fifo, leader split 4 fifo, ring split 5 fifo, ring conf 5 fifo, ring split 6 fifo, ring conf 6 fifo, ring split 7 fifo, ring conf 7 fifo, ring conf 8 fifo, ring split 8 fifo.

skipped: a smaller size had reached its limit: leader split 5 fifo, leader silent 5 fifo, leader conf 5 fifo, leader split 6 fifo, leader conf 6 fifo, leader silent 6 fifo, leader conf 7 fifo, leader silent 7 fifo, leader split 7 fifo, leader silent 8 fifo, leader conf 8 fifo, leader split 8 fifo, leader split 9 fifo, leader conf 9 fifo, leader silent 9 fifo, ring conf 10 fifo.

left out by the table rules (memory): ring split 9 fifo, ring conf 9 fifo.

cross offline, Java implementation against the Rust specification

cross offline, Java implementation against the Rust specification: pair **leader-conf**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	4	6	conforms	3.2	156	10 min, 250 GB	0.1	3.1	0.1	merge	brain01, 2026-11-03

cross offline, Java implementation against the Rust specification: pair **ring-conf**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	4	4	conforms	3.3	161	10 min, 16 GB	0.1	3.3	0.0	in-memory	brain01, 2026-11-02

cross offline, Java implementation against the Rust specification: pair **leader-silent**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	4	3	1	violation	2.9	170	10 min, 250 GB	0.1	2.9	0.0	merge	brain01, 2026-11-03

cross offline, Java implementation against the Rust specification: pair **ring-split**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	4	6	3	violation	3.1	166	10 min, 16 GB	0.1	3.1	0.0	in-memory	brain01, 2026-11-02

Not run (cross offline, Java implementation against the Rust specification). refused (an input missing at run time): leader split 2 fifo, leader conf 3 fifo, ring conf 3 fifo, leader silent 3 fifo, ring split 3 fifo, leader split 3 fifo, ring split 4 fifo, ring conf 4 fifo.

planned but never scheduled (grid points beyond the ladder): ring split 9 fifo, ring conf 9 fifo, ring conf 10 fifo, ring split 10 fifo, ring conf 11 fifo, ring split 11 fifo.

cross offline, Rust implementation against the Java specification

cross offline, Rust implementation against the Java specification: pair **leader-conf**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	4	6	conforms	3.1	155	10 min, 16 GB	3.1	0.1	0.0	in-memory	brain01, 2026-11-01
3	18	588	conforms	3.4	170	10 min, 16 GB	3.4	0.1	0.0	in-memory	brain01, 2026-11-01

cross offline, Rust implementation against the Java specification: pair **ring-conf**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	4	4	conforms	3.1	155	10 min, 16 GB	3.1	0.0	0.0	in-memory	brain01, 2026-11-01
3	18	24	conforms	3.4	170	10 min, 16 GB	3.4	0.0	0.0	in-memory	brain01, 2026-11-01
4	96	192	conforms	3.8	156	10 min, 16 GB	3.8	0.1	0.0	in-memory	brain01, 2026-11-01

cross offline, Rust implementation against the Java specification: pair **leader-silent**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	4	3	1	violation	3.1	155	10 min, 16 GB	3.1	0.0	0.0	in-memory	brain01, 2026-11-01
3	18	196	2	violation	3.4	170	10 min, 16 GB	3.4	0.1	0.0	in-memory	brain01, 2026-11-01

cross offline, Rust implementation against the Java specification: pair **leader-split**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	4	9	3	violation	3.1	155	10 min, 16 GB	3.1	0.1	0.0	in-memory	brain01, 2026-11-01
3	18	882	9	violation	3.4	170	10 min, 16 GB	3.4	0.2	0.0	in-memory	brain01, 2026-11-01

cross offline, Rust implementation against the Java specification: pair **ring-split**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	4	6	3	violation	3.1	155	10 min, 16 GB	3.1	0.0	0.0	in-memory	brain01, 2026-11-01
3	18	28	7	violation	3.4	170	10 min, 16 GB	3.4	0.1	0.0	in-memory	brain01, 2026-11-01
4	96	120	15	violation	3.8	156	10 min, 16 GB	3.8	0.1	0.0	in-memory	brain01, 2026-11-01

Not run (cross offline, Rust implementation against the Java specification). planned but never scheduled (grid points beyond the ladder): ring split 9 fifo, ring conf 9 fifo, ring conf 10 fifo, ring split 10 fifo, ring conf 11 fifo, ring split 11 fifo.

cross online, Java implementation (JMC) against the Rust specification export

cross online, Java implementation (JMC) against the Rust specification export: pair **leader-conf**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	4	6	conforms	0.7	174	1 h, 250 GB		brain05, 2026-11-02
3	18	588	conforms	3.0	256	1 h, 250 GB		brain05, 2026-11-02

cross online, Java implementation (JMC) against the Rust specification export: pair **ring-conf**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	4	4	conforms	0.6	152	1 h, 250 GB		brain05, 2026-11-02
3	18	24	conforms	0.9	150	1 h, 250 GB		brain05, 2026-11-02
4	96	192	conforms	1.9	279	1 h, 250 GB		brain05, 2026-11-02

cross online, Java implementation (JMC) against the Rust specification export: pair **leader-silent**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2	4	3	3	violation	0.6	148	1 h, 250 GB	44	brain05, 2026-11-02
3	18	196	196	violation	1.4	184	1 h, 250 GB	56	brain05, 2026-11-02

cross online, Java implementation (JMC) against the Rust specification export: pair **leader-split**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2	4	9	9	violation	0.7	153	1 h, 250 GB	48	brain05, 2026-11-02
3	18	882	882	violation	3.8	290	1 h, 250 GB	66	brain05, 2026-11-02

cross online, Java implementation (JMC) against the Rust specification export: pair **ring-split**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2	4	6	6	violation	0.7	152	1 h, 250 GB	62	brain05, 2026-11-02
3	18	28	28	violation	0.8	150	1 h, 250 GB	54	brain05, 2026-11-02
4	96	120	120	violation	1.4	207	1 h, 250 GB	67	brain05, 2026-11-02

Not run (cross online, Java implementation (JMC) against the Rust specification export). planned but never scheduled (grid points beyond the ladder): leader conf 2 fifo, leader split 2 fifo, ring split 2 fifo, leader silent 2 fifo, ring conf 2 fifo, leader split 3 fifo, leader conf 3 fifo, ring conf 3 fifo, leader silent 3 fifo, ring split 3 fifo, ring split 4 fifo, ring conf 4 fifo, ring split 9 fifo, ring conf 9 fifo, ring conf 10 fifo, ring split 10 fifo, ring conf 11 fifo, ring split 11 fifo.

cross online, Rust implementation (TraceForge) against the Java specification export

cross online, Rust implementation (TraceForge) against the Java specification export: pair **leader-conf**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	4	6	conforms	0.0	4	1 h, 250 GB		brain04, 2026-11-01
3	18	588	conforms	0.1	4	1 h, 250 GB		brain04, 2026-11-01

cross online, Rust implementation (TraceForge) against the Java specification export: pair **ring-conf**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	4	4	conforms	0.0	4	1 h, 250 GB		brain04, 2026-11-01
3	18	24	conforms	0.0	4	1 h, 250 GB		brain04, 2026-11-01
4	96	192	conforms	0.1	4	1 h, 250 GB		brain04, 2026-11-01

cross online, Rust implementation (TraceForge) against the Java specification export: pair **leader-silent**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2	4	3	3	violation	0.0	4	1 h, 250 GB	2	brain04, 2026-11-01
3	18	196	196	violation	0.0	4	1 h, 250 GB	2	brain04, 2026-11-01

cross online, Rust implementation (TraceForge) against the Java specification export: pair **leader-split**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2	4	9	9	violation	0.0	4	1 h, 250 GB	6	brain04, 2026-11-01
3	18	882	882	violation	0.1	4	1 h, 250 GB	2	brain04, 2026-11-01

cross online, Rust implementation (TraceForge) against the Java specification export: pair **ring-split**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2	4	6	6	violation	0.0	4	1 h, 250 GB	2	brain04, 2026-11-01
3	18	28	28	violation	0.0	4	1 h, 250 GB	2	brain04, 2026-11-01
4	96	120	120	violation	0.0	4	1 h, 250 GB	2	brain04, 2026-11-01

Not run (cross online, Rust implementation (TraceForge) against the Java specification export). left out by the table rules (ok): leader conf 2 fifo, leader split 2 fifo, ring split 2 fifo, leader silent 2 fifo, ring conf 2 fifo, leader split 3 fifo, leader conf 3 fifo, ring conf 3 fifo, leader silent 3 fifo, ring split 3 fifo, ring split 4 fifo, ring conf 4 fifo.

planned but never scheduled (grid points beyond the ladder): ring split 9 fifo, ring conf 9 fifo, ring conf 10 fifo, ring split 10 fifo, ring conf 11 fifo, ring split 11 fifo.